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CHM 102 — Past Questions & Materials
by Renance

Course: CHM 102
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 100

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Paper 1 of 2 — CHM 102

Level 100 • Second Semester • PastQuestion

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INTRODUCTION, ISOLATION AND Puriification

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Paper 2 of 2 — CHM 102 Material

Level 100 • Second Semester • Handout

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CHM 102 — Organic Chemistry (A)

The Chemistry of Nitriles, Aldehydes, Ketones, Carboxylic acids & derivatives, ethers, amines, alcohols and derivatives (Preparation, reactions, uses and classification) — Tuesday 12:00 to 2:00 pm (CCOS), Monday 10:00 am to 12:00 pm (COT)

1. Chemistry of Nitriles

Nitriles (Cyanides) are organic compounds containing the cyano group ($\text{C}\equiv\text{N}$) attached to a carbon atom. They can be classified based on the structure and substitution patterns. They are ~~soluble~~ (simple/small nitrile which form hydrogen bond with water) and ~~not~~ very soluble in organic solvents. They are ~~very~~ stable in normal condition. ^{Aldehydes, ketones, carboxylic acids, amines, alcohols and derivatives}

Classification

- (i) Simple Nitriles: They contain cyano group attached to the a single carbon atom. eg. Acetonitrile ($\text{C}_2\text{H}_3\text{N}$)
- (ii) Substituted Nitriles: These contain additional groups attached to the nitrile group. Example Benzonitrile ($\text{C}_6\text{H}_5\text{CN}$) and 4-nitrobenzonitrile ($\text{C}_6\text{H}_4\text{NO}_2\text{CN}$)
- (iii) Aliphatic Nitriles: These are nitriles with a cyano group attached to an aliphatic chain. eg. Propionitrile ($\text{CH}_3\text{CH}_2\text{CN}$).
- (iv) Cyclic nitrile: — Are nitriles where the cyano group is part of a ring structure. eg. Pyridine-3-Carbonitrile. Other forms of classification include:
 - i) By functional group:
 - (a) Aliphatic nitriles contain a cyano group attached to an aliphatic (non-aromatic) carbon chain. example ethyl cyanide (propionitrile, $\text{CH}_3\text{CH}_2\text{CN}$), methyl cyanide (Acetonitrile CH_3CN).
 - (b) Aromatic nitrile contain a cyano group attached to an aromatic ring eg. Benzonitrile ($\text{C}_6\text{H}_5\text{CN}$), Toluonitrile ($\text{C}_6\text{H}_5\text{CH}_2\text{CN}$).
 - ii) By substitution patterns:
 - (a) mononitriles contain a single cyano group eg. acetonitrile (CH_3CN), benzonitrile ($\text{C}_6\text{H}_5\text{CN}$).
 - (b) Dinitriles contain two cyano groups eg. Malononitrile ($\text{NC}-\text{CH}_2-\text{CN}$), fumaredinitrile ($\text{NC}-\text{C}=\text{C}-\text{CN}$).

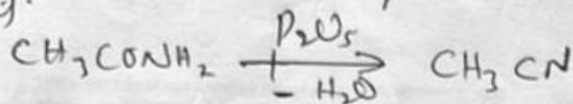
$\text{CH}_3-\text{CH}(\text{CN})$

- i) By cyclic structure (a) acyclic nitriles are linear ^{or branched} they do not form a ring eg. propionitrile ($\text{CH}_3\text{CH}_2\text{CN}$) and isobutyronitrile ($\text{C}(\text{CH}_3)_2\text{CH}_2\text{CN}$). (b) cyclic nitriles have cyano groups which are form part of the ring structure eg. pyridine-3-carbonitrile and cyclopropane-1-carbonitrile.
- ii) By Reactivity (a) primary nitriles have the cyano group directly attached to a primary carbon. eg. butyronitrile ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$). (b) secondary nitriles have the cyano group attached to a secondary carbon, often within a complex structure. Examples are some nitriles formed by specific reactions of secondary alkyl halides & tertiary nitrile eg. tert-butyl nitrile.
- iii) By molecular size (a) small nitriles have fewer carbons and simple structures. eg. Acetonitrile (CH_3CN), propionitrile ($\text{CH}_3\text{CH}_2\text{CN}$). (b) Large nitriles have complex structures or longer carbon chains eg. Dodecane nitrile ($\text{C}_{12}\text{H}_{25}\text{CN}$), bicyclic nitriles like those found in natural products.

Preparation

- i) From alkyl halides: $\text{Alkyl halide} + \text{Sodium cyanide} \rightarrow \text{nitrile}$
- $\text{Alkyl halide} + \text{Sodium cyanide} \rightarrow \text{nitrile} + \text{Sodium halide}$
- $\text{ethyl bromide} + \text{NaCN} \rightarrow \text{CH}_3\text{CH}_2\text{CN} + \text{NaBr}$
- $\text{C}_2\text{H}_5\text{Br} \rightarrow \text{ethyl cyanide}$
- ii) From Carbonyl Compounds:
- (a) Carbonyl compound + Hydroxylamine or semi carbazide $\rightarrow$ Oxime or semicarbazone + Cyanide im (Nitrile)
- $\text{CH}_3\text{COCH}_3 + \text{Hydroxylamine} \rightarrow \text{Acetone oxime} \xrightarrow[\text{-H}_2\text{O}]{\text{dehydration}} \text{Acetonitrile}$
- (b) Carboxylic acid, eg. acetic acid is reacted with thionyl chloride to form acetyl chloride which then reacts with NaCN to produce acetonitrile
- $\text{CH}_3\text{COOH} + \text{SOCl}_2 \rightarrow \text{CH}_3\text{COCl}$
- $\text{acetyl chloride} + \text{NaCN} \rightarrow \text{Acetonitrile}$

from amide. In the presence of a dehydrating agent, amides are dehydrated to form nitriles.
 eg.

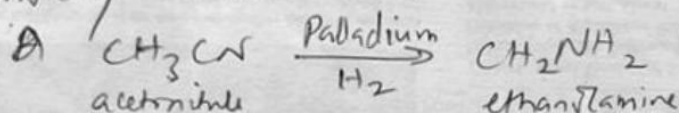


Reactions of Nitrile:

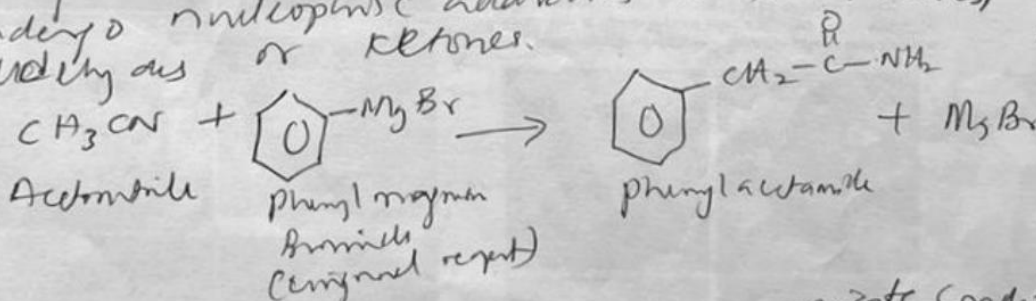
(i) Hydrolysis: - Nitriles undergo hydrolysis under acidic or alkaline conditions to form an amine or ammonia. For example, Acetonitrile hydrolyse to form acetic acid and ammonia.



(ii) Reduction: In the presence of suitable catalysts, nitrile undergoes reduction to produce the corresponding primary amine. Eg.



(iii) Nucleophilic Addition: - In the presence of Lewis strong electron donor (Lewis base), nitriles undergo nucleophilic addition to form corresponding aldehydes or ketones.



(iv) Cyclization: - In the presence of appropriate conditions and reagents, nitriles undergo cyclization to form cyclic compounds like cyclic amides. Further, ~~benzene~~ aromatic nitriles can cyclize to form heterocyclic compounds like pyridine.

Uses of nitriles

Nitriles have diverse applications in industry and research.

- i. Pharmaceuticals: Nitriles are intermediates in the synthesis of drugs and biologically active molecules.
Example: The synthesis of antihypertensive drug like nifedipine.
- ii. Solvents: - Acetonitrile is used as a solvent in extraction process as well as in chemical analysis due to its high polarity and viscosity.
- iii. Synthetic polymers: - nitriles are used in the ~~in auto~~ ~~native and industrial~~ production of synthetic polymers such as ~~acrylonitrile~~ nitrile rubber (Buna-N), which is used in the automotive and industrial applications for its resistance to oils and fuels deterioration.
- (iv) Chemical reagents: Nitriles serve as precursors in the synthesis of various chemicals, including adhesives and pesticides.

Exercises

1. Give two examples of the following: (a) aliphatic nitriles (b) aromatic nitriles (c) mononitriles (d) dinitriles (e) primary ^{nitrile} (f) secondary nitriles (g) tertiary nitriles
2. What are the differences (or similarities) between cyclic nitriles and aromatic nitrile.
3. Give two major ways for the preparation of nitriles.
4. "In the presence of appropriate conditions and reagents, nitriles undergo cyclization to form cyclic compounds." With named compounds, give a chemical equation to represent the above statement, showing appropriate catalysts and reaction conditions.
5. List the applications of nitriles in the petroleum industry.
6. Discuss the chemistry of nitriles under the heading (a) nomenclature (b) physical properties (c) derivatives

CHM 102

(B)

Aldehydes

CHEMISTRY OF ALDEHYDES, KETONES, CARBOXYLIC ACID & DERIVATIVES

Chemistry of Carbonyls

1. ALDEHYDES / ALKANALS

- General formula $R-CHO$, Carbonyls where $R = \text{alkyl or aryl}$
- Nomenclature: "al" of alkane replaced with "al", eg. methane $\rightarrow$ methanal
- Physical Classification: Simple alkanal have one R , Substituted have additional groups.

Physical properties of Alkanal:

- * Lower boiling point than corresponding alkanal
- * Higher bp than corresponding alkanes
- * dipole-dipole interaction due to $C=O$ Carbonyl group
- * Often have strong characteristic color.

Preparation of Aldehydes

$R-CH_2OH \xrightarrow{PCC} R-CHO$ (PCC), an oxidizing agent, alcohols are converted to aldehyde

$R-COCl \xrightarrow{LiAlH_4} R-CHO$ Reduction of acid chlorides: - Acid chlorides are reduced using $LiAlH_4$ to produce alkanal

$R-CN \xrightarrow{H_2O} R-CHO$ Hydrolysis of nitriles: - Acid Dinitrile acids or bases are reacted with nitriles, followed by reduction to yield alkanal

$R-CH=CH-R' \xrightarrow{O_3} R-CHO + R'-CHO$ Oxidation of aldehyde precursors like alkanes via ozonolysis followed by reduction yields alkanal.

$R-CH_3 \xrightarrow{Cl_2, UV} R-CHO$ Wohl-Ziegler oxidation: - This reaction involves the oxidation of alkanes using chlorine in the presence of UV light to form aldehydes.

Reactions of Aldehydes: Aldehydes can be oxidized to carboxylic acid using oxidizing agents like $KMnO_4$ or chromate.

* Reduction: Aldehydes can be reduced to alcohols in the presence of reducing agents (1)

ice Sodium bisulphide (NaHSO_3) or lithium aluminium hydride (LiAlH_4).

- Nucleophilic Addition reactions such as the addition of hydride ions from reducing agents or nucleophile like cyanide ions (CN^-) to form cyanohydrins
- * Reaction with alcohols to form hemiacetals or acetals

Uses of Aldehydes

- * Formaldehyde (methanal) is used in the production of resins (e.g. urea-formaldehyde), disinfectants, and embalming fluids. Also used in the manufacturing of plastics and textiles
- * Acetaldehyde (ethanal) is utilized in the synthesis of acetic acids and other chemicals like flavourings and perfumes
- * Benzaldehyde is used as flavouring agent (almond flavour) and in the production of dyes and pharmaceuticals

2. KETONES / ALKANONES

- General formula $\rightarrow \text{R}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{R}'$ where R & R' are alkyl or aryl

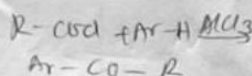
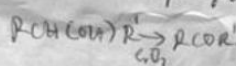
- Nomenclature $\rightarrow$ "ket" $\rightarrow$ "one"

- Classification $\rightarrow$ Simple and substituted

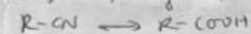
- Physical properties of alkanone

- * Like alkanals, they have higher bp than alkanes but lower than alkanols of similar mol. wt.
- * Less reactive than alkanals due to lack of hydrogen on the carbonyl carbon.

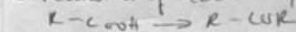
- Preparation of Alkanones



(i) Hydrolysis of nitrile



(ii) Reduction of acid



* Oxidation of secondary alcohols yield correspondingly Alkanone

* Friedel-Crafts Acylation: An acyl chloride ($\text{R}-\text{COCl}$) reacts with an aromatic compound in the presence of a Lewis acid catalyst like AlCl_3 to produce alkanone

* Hydrolysis of nitriles: Nitriles can be hydrolysed and subsequently reduced to form corresponding alkanone

* Kolbe Electrolysis: Electrolysis of sodium or potassium salts of carboxylic acids yield ketone $2\text{R}-\text{COONa} \xrightarrow{\text{electrolysis}} \text{R}-\text{COR}'$

3. CARBOXYLIC ACIDS / ALKANOIC ACIDS $\text{C}=\text{O}$ Carbonyl

— General formula $\text{R}-\text{COOH}$ where $\text{R} = \text{alkyl or aryl}$

— Nomenclature: "e" = "oic"

— Classification: Based on the nature of constituents: simple and substituted.

* Based on the number of carboxyl groups

— mono (e.g. acetic), dicarboxylic (e.g. oxalic, succinic, tetracarboxylic e.g. succinic acid).

— Physical properties

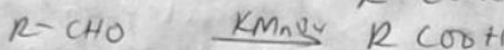
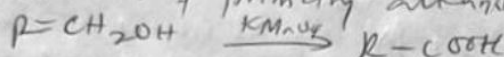
* higher boiling points than aldehydes & ketones due to strong hydrogen bonding.

* more soluble in water compared to aldehydes & ketones of similar mol

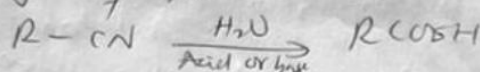
* dipole-dipole interactions due to $\text{C}=\text{O}$ and OH

— Preparation

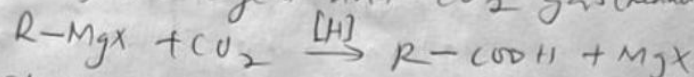
* Oxidation of primary alkanols and ^{aldehydes}



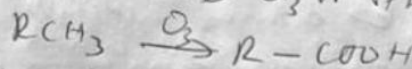
* Hydrolysis of Nitriles



* From Grignard reagent and CO_2 gas (Reductive gas)

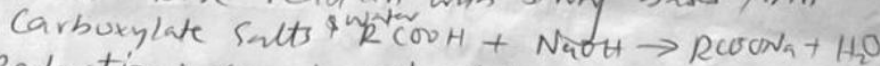


* Oxidation of Alkanes (Controlled oxidation of alkanes with ozone O_3 or other oxidizers).



— Reaction of Alkanoic Acids

* Acid-Base reaction with strong bases form Carboxylate salts ^{to}



* Reduction:— Can be reduced to primary alkanols or aldehydes. LiAlH_4 can reduce alkanoic acid to primary alkanols

* Formation of Derivatives:— ^{can} form various derivatives such as esters, amides, and ~~by~~ anhydrides through condensation reactions.

* Decarboxylation:— Carboxylic acids can lose a CO_2 molecule upon heating with soda-lime or certain other reagents, resulting in the corresponding hydrocarbon.

(4)

- uses of Carboxylic Acids
- * Acetic Acids: Used in the production of vinegar, Synthetic Fibers (eg acetate), and as a chemical reagent in various industrial process
 - * Formic Acid: Utilized in leather production, as a preservative and in agriculture.
 - * Citric Acids: Widely used as a flavouring agent in food and beverages, in pharma contents and as a chelating agent in cleaning products.

~~4. Deriva~~

4. DERIVATIVES OF CARBOXYLIC ACIDS

— This include Esters, Amides, Anhydrides and Acid chlorides

— General ~~Formula~~ Summary: ^{where!} R = alkyl or aryl

Parameter	Esters	Amides	Anhydrides	Acid chloride
General formula	$R-COOR'$ $R-\overset{\overset{O}{\parallel}}{C}-O-R'$	$R-CONH_2$ / $R-\overset{\overset{O}{\parallel}}{C}-NH_2$ $R-CONHR'$ / $R-\overset{\overset{O}{\parallel}}{C}-NH-R'$ $R-CONR'R''$ / $R-\overset{\overset{O}{\parallel}}{C}-N(R')R''$ $R-CONR'R''$ / $R-\overset{\overset{O}{\parallel}}{C}-N(R')R''$ $R-CONR'R''$ / $R-\overset{\overset{O}{\parallel}}{C}-N(R')R''$	$R-CO-O-CO-R'$ $R'-\overset{\overset{O}{\parallel}}{C}-O-\overset{\overset{O}{\parallel}}{C}-R''$	$R-CO-Cl$ $R-\overset{\overset{O}{\parallel}}{C}-Cl$
2. Classification	Simple: From single Carboxylic acid and alcohol eg methyl methanoate Complex: From a simple Carboxylic acid and a more complex alcohol or phenol eg. Lipids, methyl Salicylate	$R-\overset{\overset{O}{\parallel}}{C}-NH_2$ 1° Acetamide $R-\overset{\overset{O}{\parallel}}{C}-NH-CH_3$ 2° N-methyl acetamide $R-\overset{\overset{O}{\parallel}}{C}-N(CH_3)_2$ 3° N,N-dimethylacetamide Alkanolic acid and ammonia or amine	Symmetrical anhydride classed same way as the carboxylic acid eg. Acetic anhydride Asymmetrical anhydride or mixed anhydride: From two different carboxylic acids eg. Acetic propanoic acid	Carboxylic acid with thionyl chloride ($SOCl_2$) or Oxalyl chloride ($COCl_2$)
3. Formation	Alkanolic acid and alcohol	Alkanolic acid and ammonia or amine	two alkanolic acids of the same or different kinds, via condensation with a loss of water	Carboxylic acid with thionyl chloride ($SOCl_2$) or Oxalyl chloride ($COCl_2$)

(5)

	Esters	Amide	Anhydride	Acid chloride
Reaction:	① Fischer esterification: $R\text{COOH} + R'\text{OH} \rightarrow R\text{COOR}' + \text{H}_2\text{O}$ ② Transesterification: esters react with alkanes to form different esters. $R\text{COOR}'' + R'\text{OH} \rightarrow R\text{COOR}' + R''\text{OH}$	① Direct reaction of Carboxylic acids with amines or ammonia: $R\text{COOH} + \text{RNH}_2 \xrightarrow{\text{DCC}} R\text{CO-NH-R}'$ ② With ammonia: $R\text{COOH} + \text{NH}_3 \rightarrow R\text{CO-NH}_2 + \text{H}_2\text{O}$	Treatment of Carboxylic acids with acid chlorides, or other anhydrides: $R\text{COOH} + R'\text{COOH} \xrightarrow{\text{acid chloride}} R\text{CO-O-CO-R}'$	$R\text{COOH} + \text{SOCl}_2 \rightarrow R\text{COCl} + \text{SO}_2 + \text{HCl}$
in	① Hydrolysis: Can be hydrolysed back into Carboxylic acid and alcohol. ② Saponification: can form soap in the presence of strong alkali like KOH or NaOH to form soap and water (or alcohol).	Hydrolysis: Can be hydrolysed to form carboxylic acid and amines.	① With alkanes to form esters ② With amines to form amides	Highly reactive and can form esters, amides and other derivatives depending on the condition
	① Ethyl acetate: used as a solvent in coatings, adhesives, and as a flavouring agent. ② Methyl Salicylate: utilized (Wintergreen oil), used in medicinal (cannabis) and flavouring applications (in chewing gum).	① Acetamide: used as a solvent in the production of pharmaceuticals & dyes. ② N,N-Dimethylacetamide: As a solvent in polymer processing and as a reagent in organic synthesis.	① Acetic Anhydride: used in the manufacture of acetylated compounds such as aspirin, and in the production of cellulose acetate. ② Mixed anhydrides: used in various chemical synthesis and pharmaceutical applications.	① Acetyl chloride: used in the synthesis of acetyl derivative and in the production of pharmaceuticals. ② Benzoyl chloride: used in the manufacture of dyes, pharmaceuticals, and as a reagent in organic synthesis.

⑥